

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 2 Term 2 2021

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 65 minutes
- Write using black pen
- Calculators approved by NESA may be used
- In Questions 6 – 11, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination
- Students are permitted to refer to one double-sided A4 page of handwritten topic summaries which they prepared prior this assessment

Total marks – 52

Section I Pages 2 – 4

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Answer on the Objective Response Answer Sheet provided

Section II Pages 5 – 14

47 marks

Attempt Questions 6 -11

Allow about 57 minutes for this section

Write your answers in the space provided.

Question	1-5	6	7	8	9	10	11	Total
Total	/5	/9	/7	/8	/7	/8	/8	/52

This assessment task constitutes 30% of the Preliminary Higher School Certificate Course School Assessment

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

1 Given $f(x) = x + 5$ and $g(x) = x - 2$, what is the domain of $y = \frac{f(x)}{g(x)}$?

(A) $(2, \infty)$

(B) $(-\infty, -5) \cup (-5, \infty)$

(C) $(-\infty, 2] \cup [2, \infty)$

(D) $(-\infty, 2) \cup (2, \infty)$

2 Which equation is a many to one relation?

(A) $y = \pm\sqrt{9 - x^2}$

(B) $y = \sqrt{9 - x^2}$

(C) $x = \sqrt{9 - y^2}$

(D) $x = \pm\sqrt{9 - y^2}$

3 Which of the following is the vertex of the parabola $y = 2x^2 + 8x - 11$?

(A) $(-2, -19)$

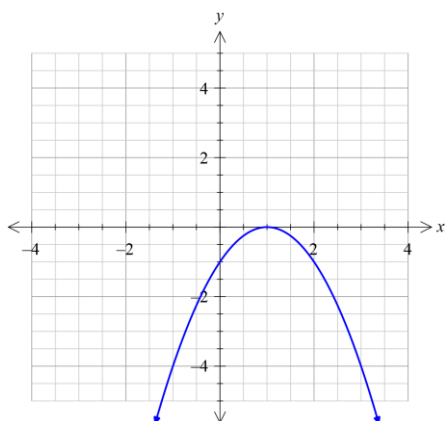
(B) $(2, 13)$

(C) $(-2, 13)$

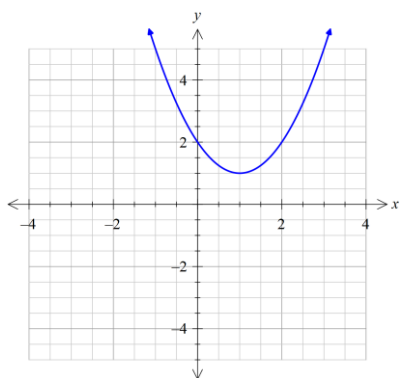
(D) $(2, 19)$

- 4 Which graph below represents a quadratic function in the form $y = ax^2 + bx + c$, where $\Delta > 0$ and $a > 0$?

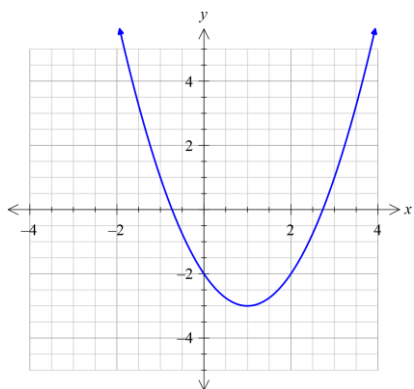
(A)



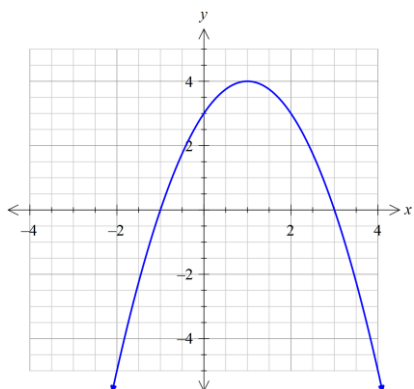
(B)



(C)

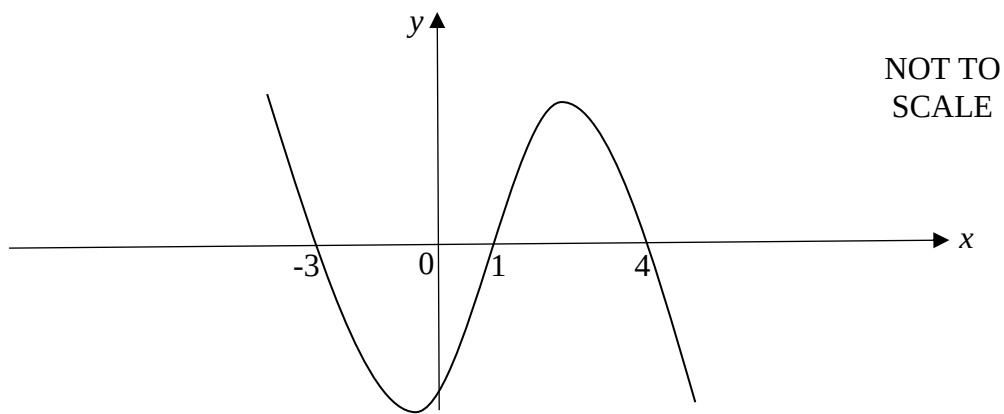


(D)



5

Which of the following is the equation for the function below?



- (A) $y = (x - 1)(x + 3)(x - 4)$
- (B) $y = (1 - x)(x + 3)(x - 4)$
- (C) $y = (1 - x)(x + 3)(4 - x)$
- (D) $y = (x + 1)(x - 3)(x + 4)$

End of Section I

Section II

47 marks

Attempt Questions 6-11

Question 6 (9 marks)

- (a) Convert 24° to radians, expressing your answer in terms of π in simplified form. 1

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- (b) What is the exact value of $\sin 240^\circ$? 2

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- (c) Is the function $f(x) = \frac{2x^2}{x^3 + 1}$ odd, even or neither? Show all working. 2

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Question 6 continues on next page

Question 6 (continued)

(d) A function is defined by

$$g(x) = \begin{cases} -x^3 & \text{for } x < -1 \\ x & \text{for } -1 \leq x \leq 1 \\ x^2 & \text{for } x > 1 \end{cases}$$

(i) Evaluate $g(-2) + g(0) + g(2)$.

1

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(ii) Sketch $g(x)$, showing all important features.

3

End of Question 6

Question 7 (7 marks)

- (a) Find the point(s) of intersection of the line $x - y - 4 = 0$ and hyperbola $y = \frac{5}{x}$. **3**

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- (b) Find the angle $3x + 4y + 9 = 0$ makes with the x -axis in the positive direction. Express your answer correct to the nearest minute. **2**

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- (c) The frequency, in chirps/minute, of a cricket's chirp is directly proportional to the temperature, in °C, of its surroundings. On a day where the temperature was 30°C, the frequency of chirps was 280 chirps/minute.

- (i) Find a formula for C in terms of T , where C is the frequency of chirps and T is the surrounding temperature. **1**

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- (ii) Use the formula found in part (i) to calculate the temperature of the surrounding environment when the chirping reaches 350 chirps/minute. Answer correct to the nearest degree. **1**

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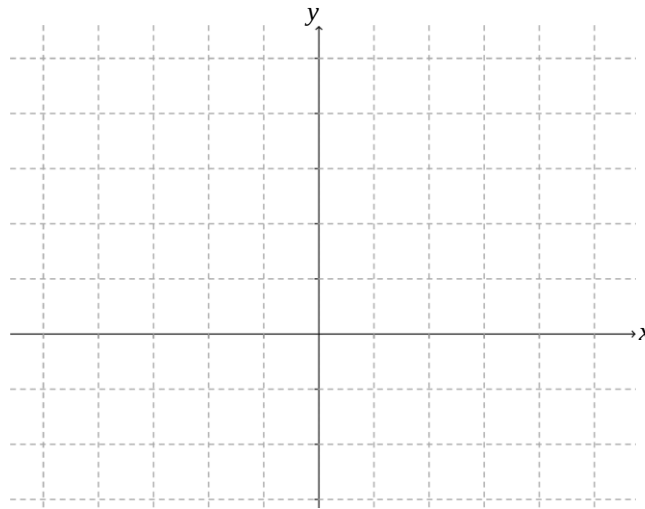
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Question 8 (8 marks)

- (a) (i) Sketch $y = |x| - 1$ and $y = 4$ on the number plane below, showing the x and y -intercepts.



2

- (ii) Using your graph, or otherwise, solve $|x| - 1 = 4$.

2

.....

- (b) A sheet of cardboard, measuring 30cm by 24cm, has equal squares of side length x cm cut out of its corners. The cut-out sides are then folded up to form an open box.

- (i) Write down an expression for the area of the base of the box in terms of x .

1

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- (ii) Find the dimensions of the base of the box given that its area is 216cm^2 .

3

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Question 9 (7 marks)

(a) If $f(x) = \frac{2}{x}$, show that $f(x+2) - f(x) = -\frac{4}{x^2 + 2x}$.

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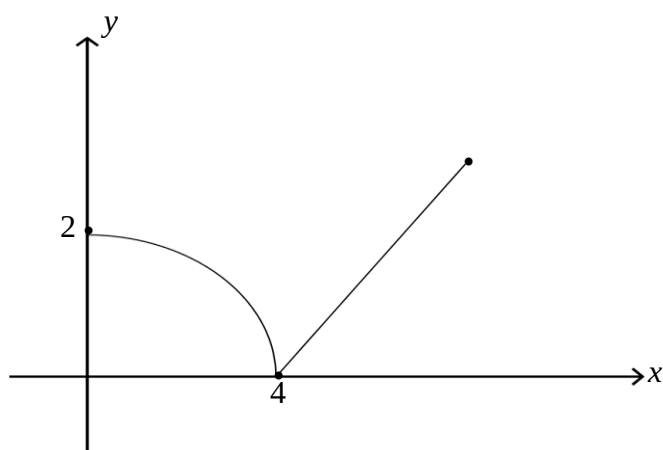
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(b) The diagram below shows the graph of $y = f(x)$ with endpoints $(0,2)$ and $(7,3)$.



NOT TO
SCALE

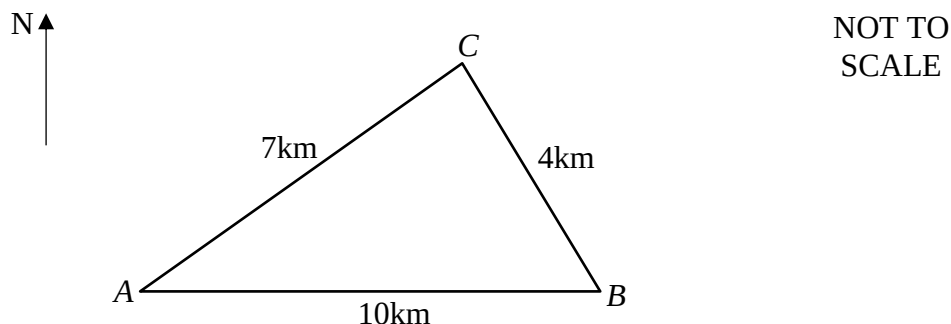
Sketch $y = f(x+1) + 3$ in the space below, showing the new endpoints.

2

Question 9 continues on next page

Question 9 (continued)

- (c) In the diagram below, building B is 10km due east of building A . Building C is 7km from building A and 4km from building B .



- (i) Find the size of $\angle ABC$, correct to the nearest degree.

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- (ii) Find the bearing of building B from building C , correct to the nearest degree.

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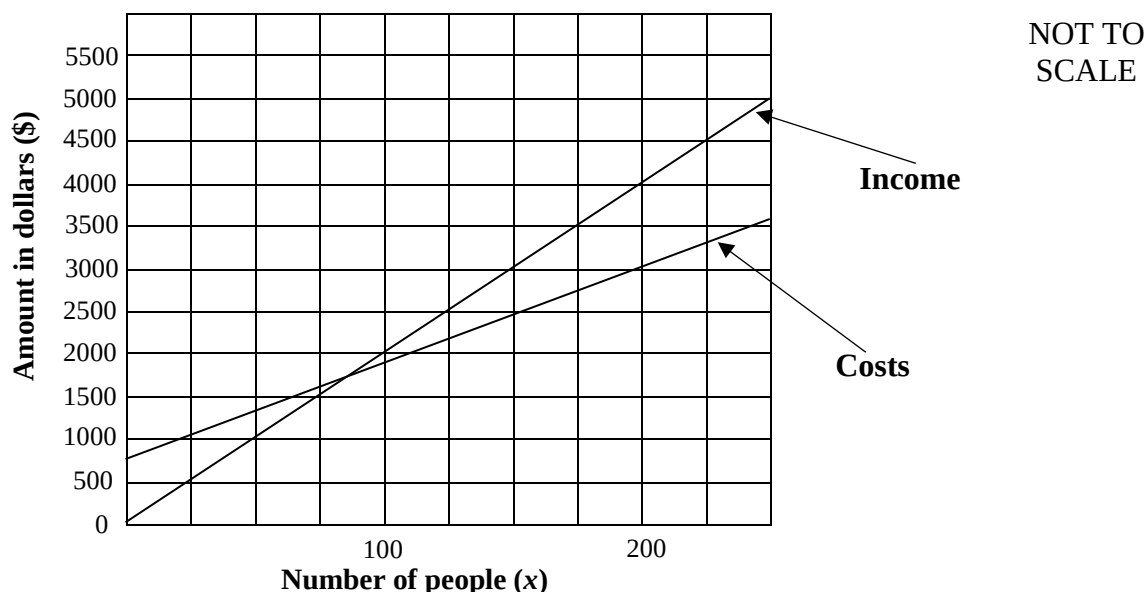
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Question 10 (8 marks)

- (a) Susan and Nick are planning a fund-raising dance. They hire a hall for \$400 and a band for \$300. Refreshments will cost \$12 per person. They set the entry ticket at \$20 per person. The graph below shows planned income and costs for the dance.



- (i) Write a formula for the cost, C , of running the dance for x people. 1

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- (ii) Estimate the minimum number of people needed at the dance to cover the costs. 1

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- (iii) How much profit will be made if 150 people attend the dance? 1

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- (iv) Susan and Nick plan to sell 200 tickets. They want to make a profit of \$1 500. What should be the price of a ticket assuming all 200 tickets will be sold? 2

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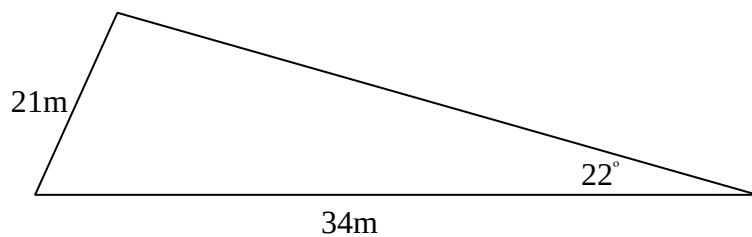
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Question 10 continues on next page

- (b) Calculate the value of θ , correct to the nearest minute.

3



NOT TO
SCALE

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Question 11 (8 marks)

- (a) Solve for x : $\cos x = -\frac{\sqrt{2}}{2}$ for $0^\circ \leq x \leq 360^\circ$.

2

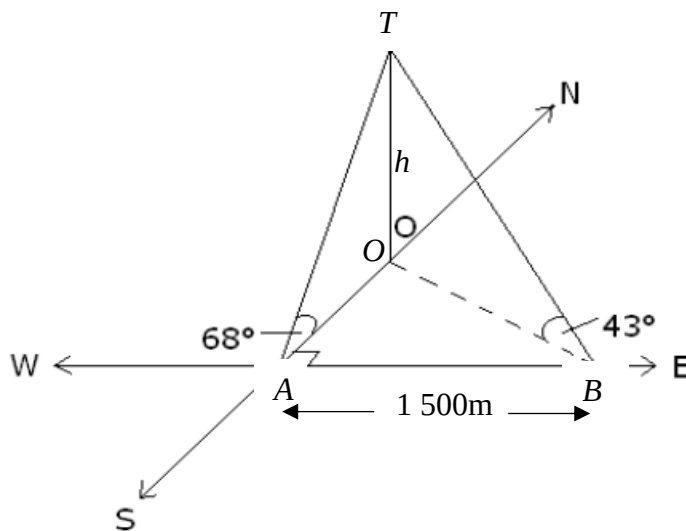
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- (b) A lighthouse of height h metres, is observed from two points, A and B , in the same horizontal plane as the base of the lighthouse. From point A , due south of the lighthouse, the angle of elevation to the top of the lighthouse is 68° . From point B , due east of point A , the angle of elevation to the top of the lighthouse is 43° . Points A and B are 1 500 metres apart.



NOT TO
SCALE

- (i) Show that $OA = h \tan 22^\circ$

2

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Question 11 continues on next page

Question 11 (continued)

- (ii) Show that $h = \frac{1\,500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}$ **3**

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- (iii) Find the height of the tower correct to the nearest metre. **1**

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END OF PAPER

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Mathematics Advanced

Year 11 Assessment 2
Term 2 2021

STUDENT NUMBER: _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 2 Term 2 2021

STUDENT NUMBER: SOLUTIONS and markers feedback

General Instructions

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(A) $(2, \infty)$

(B) $(-\infty, -5) \cup (-5, \infty)$

(C) $(-\infty, 2] \cup [2, \infty)$

☒ (D) $(-\infty, 2) \cup (2, \infty)$

$$y = \frac{x+5}{x-2}$$

D
B
A
C
B

- 2 Which equation is a many to one relation?

(A) $y = \pm\sqrt{9 - x^2}$

☒ (B) $y = \sqrt{9 - x^2}$

(C) $x = \sqrt{9 - y^2}$

(D) $x = \pm\sqrt{9 - y^2}$

- 3 Which of the following is the vertex of the parabola $y = 2x^2 + 8x - 11$?

☒ (A) $(-2, -19)$

(B) $(2, 13)$

(C) $(-2, 13)$

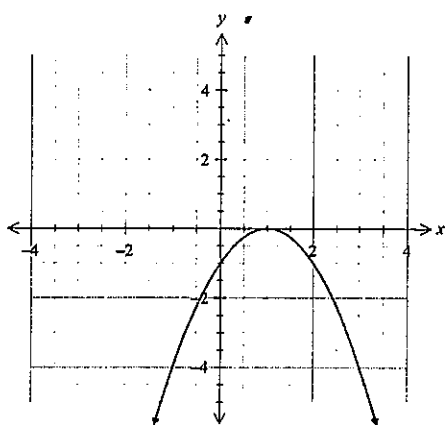
(D) $(2, 19)$

$$y = 2(x^2 + 4x + 4) - 11 - 8$$

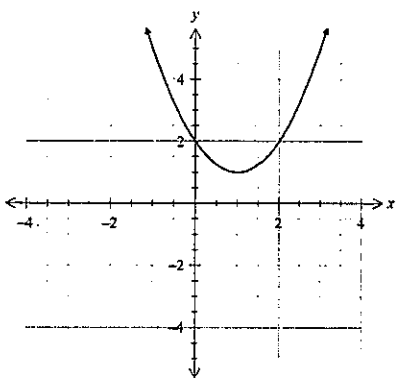
$$y = 2(x+2)^2 - 19$$

4 Which graph below represents a quadratic function in the form $y = ax^2 + bx + c$, where $\Delta > 0$ and $a > 0$?

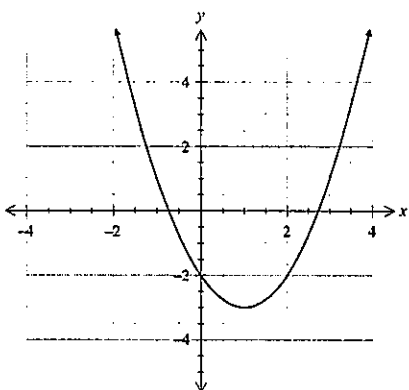
(A)



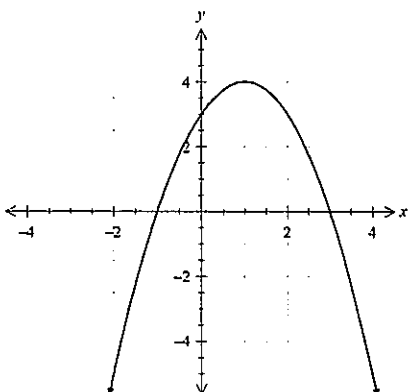
(B)



(C)

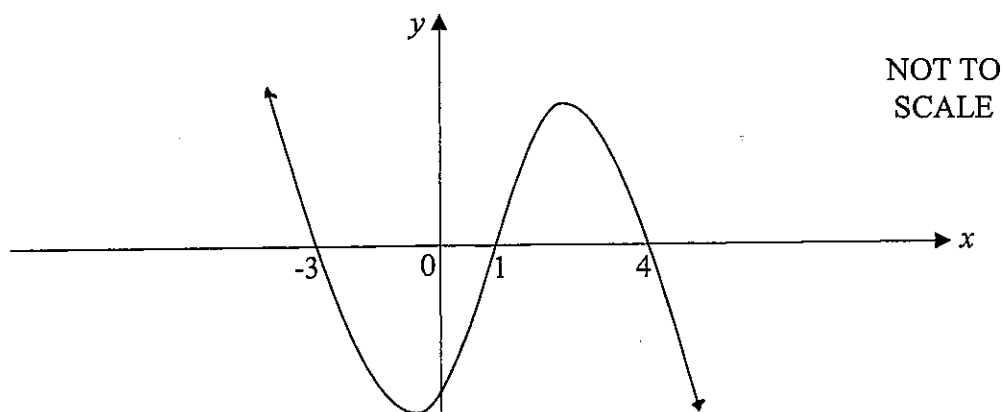


(D)



5

Which of the following is the equation for the function below?



(A) $y = (x - 1)(x + 3)(x - 4)$

(B) $y = (1 - x)(x + 3)(x - 4)$

(C) $y = (1 - x)(x + 3)(4 - x)$

(D) $y = (x + 1)(x - 3)(x + 4)$

End of Section I

Section II

47 marks

Attempt Questions 6-11

Question 6 (9 marks)

- (a) Convert 24° to radians, expressing your answer in terms of π in simplified form. 1

$$\frac{24\pi}{180}$$
$$= \frac{2\pi}{15}$$

Done well by most students

- (b) What is the exact value of $\sin 240^\circ$? 2

$$\sin 240^\circ$$
$$= \sin(180^\circ + 60^\circ)$$
$$= -\sin 60^\circ$$
$$= -\frac{\sqrt{3}}{2}$$

Done well by most students.
A few missed the negative.

- (c) Is the function $f(x) = \frac{2x^2}{x^3+1}$ odd, even or neither? Show all working. 2

$$f(x) = \frac{2x^2}{x^3+1}$$
$$f(-x) = \frac{2(-x)^2}{(-x)^3+1}$$

MUST show the $(-x)$ being substituted

$$= \frac{2x^2}{-x^3+1}$$

not even as $f(-x) \neq f(x)$ MUST justify why it is neither

$$= \frac{-2x^2}{x^3-1}$$

not odd as $f(-x) \neq -f(x)$

$\therefore$ neither odd nor even.

A common error was stating $\frac{2x^2}{-x^3+1} = -\frac{2x^2}{x^3+1}$

$$\frac{2x^2}{-x^3+1} = \frac{2x^2}{-(x^3-1)}$$
$$= -\frac{2x^2}{x^3-1}$$

Question 6 (continued)

(c) A function is defined by

$$g(x) = \begin{cases} -x^3 & \text{for } x < -1 \\ x & \text{for } -1 \leq x \leq 1 \\ x^2 & \text{for } x > 1 \end{cases}$$

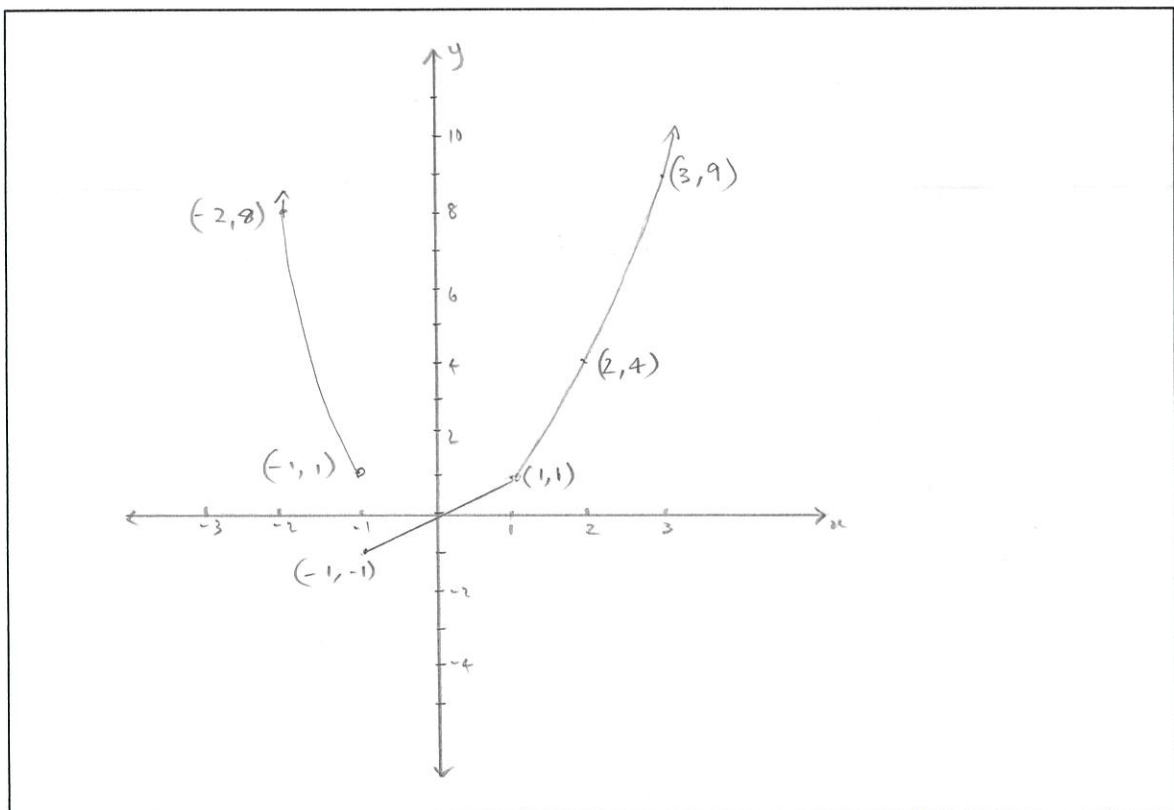
(i) Evaluate $g(-2) + g(0) + g(2)$.

1

$$\begin{aligned} & g(-2) + g(0) + g(2) \\ &= -(-2)^3 + 0 + 2^2 \\ &= 8 + 4 \\ &= 12 \end{aligned}$$

(ii) Sketch $g(x)$, showing all important features.

3



Required features

- cubic $y = -x^3$ for $x < -1$
- line $y = x$ for $-1 \leq x \leq 1$
- parabola $y = x^2$ for $x > 1$
- appropriate scale

End of Question 6

Question 7 (7 marks)

- (a) Find the point(s) of intersection of the line $x - y - 4 = 0$ and hyperbola $y = \frac{5}{x}$. 3

$$x - y - 4 = 0 \quad \text{--- (1)} \quad y = \frac{5}{x} \quad \text{--- (2)}$$

sub. (2) into (1) $x - \frac{5}{x} - 4 = 0$ *mostly well done*

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$\therefore x = 5, x = -1$ *some did not make the quad. = 0*

when $x = 5, y = \frac{5}{5} = 1$ *$\therefore$ struggled to find correct sol.*

when $x = -1, y = \frac{5}{-1} = -5$

$\therefore (5, 1) \text{ \& } (-1, -5) \text{ are pts of intersection}$

- (b) Find the angle $3x + 4y + 9 = 0$ makes with the x -axis in the positive direction. Express your answer correct to the nearest minute. 2

$$4y = -3x - 9$$

$$m = -\frac{3}{4} \quad \therefore \theta = 143^\circ 8'$$

$\tan \theta = -\frac{3}{4}$ *a lot do not know what this means*

$\therefore \theta = 143.1301024$

- (c) The frequency, in chirps/minute, of a cricket's chirp is directly proportional to the temperature, in $^\circ\text{C}$, of its surroundings. On a day where the temperature was 30°C , the frequency of chirps was 280 chirps/minute.

- (i) Find a formula for C in terms of T , where C is the frequency of chirps and T is the surrounding temperature. 1

$$C = kT \quad \therefore k = \frac{28}{3}$$

A lot got this

$$280 = k(30) \quad \therefore C = \frac{28}{3}T \quad \text{as } C = \frac{3}{28}T$$

- (ii) Use the formula found in part (i) to calculate the temperature of the surrounding environment when the chirping reaches 350 chirps/minute. Answer correct to the nearest degree. 1

$$350 = \frac{28}{3}T$$

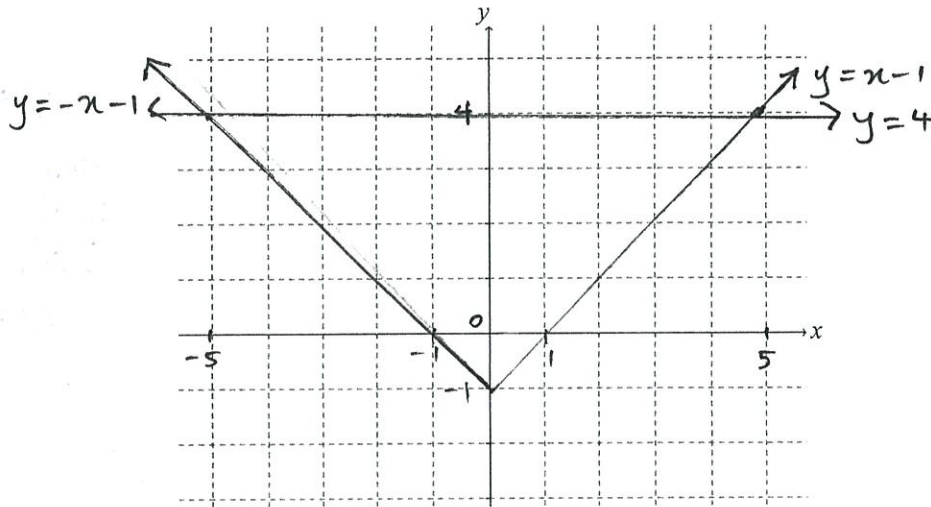
$$T = 37.5^\circ$$

$$T = 38^\circ$$

Mostly well done even when (i) was incorrect.

Question 8 (8 marks)

- (a) (i) Sketch $y = |x| - 1$ and $y = 4$ on the number plane below, showing the x and y -intercepts.



2

Many did well.
Some sketched $y = |x - 1|$ instead.
A few did not label the graph drawn.

- (ii) Using your graph, or otherwise, solve $|x| - 1 = 4$.

2

$x = \pm 5$

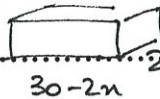
Many did well.

A few did not solve for x but gave the points of intersection which is not required.

- (b) A sheet of cardboard, measuring 30cm by 24cm, has equal squares of side length x cm cut out of its corners. The cut-out sides are then folded up to form an open box.

- (i) Write down an expression for the area of the base of the box in terms of x .

1



$A = (30 - 2x)(24 - 2x)$

Many did well.

Algebraic technical problem: common factor of 4, NOT 2.

- (ii) Find the dimensions of the base of the box given that its area is 216cm^2 .

3

$216 = (30 - 2x)(24 - 2x)$

$216 = 720 - 60x - 48x + 4x^2$

$4x^2 - 108x + 504 = 0$

$126 = 21 \times 6$

$x^2 - 27x + 126 = 0$

$(x - 21)(x - 6) = 0$

$\therefore x = 21$, $x = 6$ $\therefore x = 6$ is only soln

not possible because $x < 12$ or other explanations.

dimensions of base : $(30 - 2(6)) \times (24 - 2(6))$
 $= 18\text{cm} \times 12\text{cm}$

Question 9 (7 marks)

- (a) If $f(x) = \frac{2}{x}$, show that $f(x+2) - f(x) = -\frac{4}{x^2 + 2x}$.

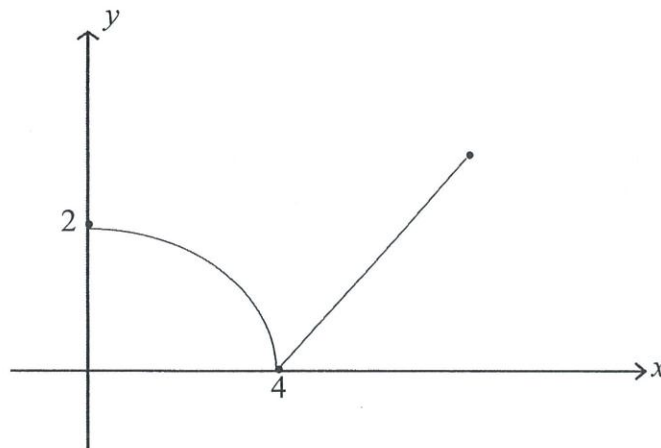
2

$$\begin{aligned}
 & f(x+2) - f(x) \\
 &= \frac{2}{x+2} - \frac{2}{x} \\
 &= \frac{2x - 2(x+2)}{x(x+2)} \\
 &= \frac{2x - 2x - 4}{x^2 + 2x} \\
 &= \frac{-4}{x^2 + 2x}
 \end{aligned}$$

It is a show question, need to show both of these lines.

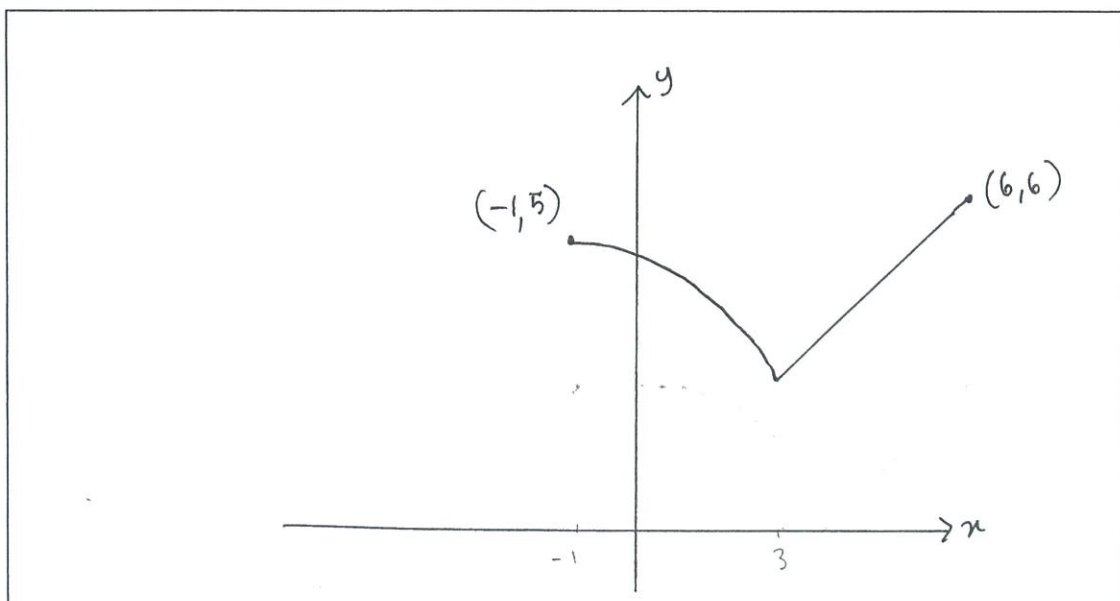
many still cannot expand $-2(x+2) \neq -2x+4$

- (b) The diagram below shows the graph of $y = f(x)$ with endpoints $(0,2)$ and $(7,3)$.



Sketch $y = f(x+1) + 3$ in the space below, showing the new endpoints.

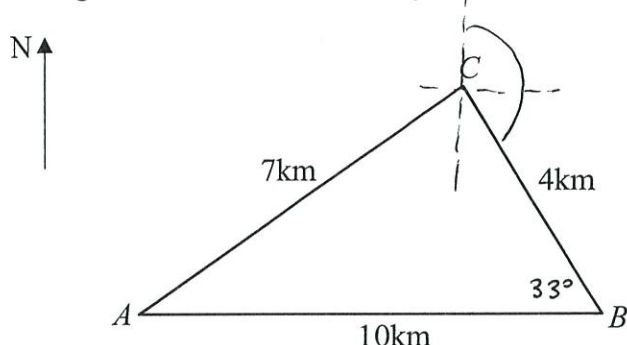
2



Question 9 continues on next page

Question 9 (continued)

- (c) In the diagram below, building B is 10km due east of building A . Building C is 7km from building A and 4km from building B .



NOT TO
SCALE

- (i) Find the size of $\angle ABC$, correct to the nearest degree.

2

let $\angle ABC = \theta$

$$\cos \theta = \frac{10^2 + 4^2 - 7^2}{2(10)(4)}$$

Done well.

$$\theta = 33.1229$$

$$\theta = 33^\circ$$

Some did not round to nearest degree.
This error was ignored.

- (ii) Find the bearing of building B from building C , correct to the nearest degree.

1

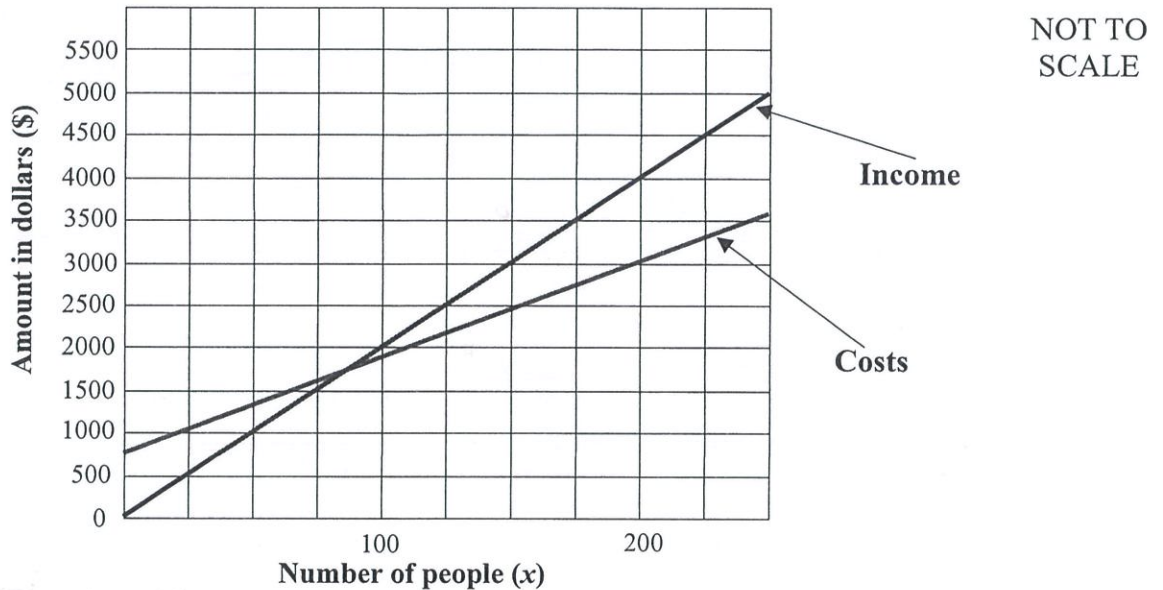
$$\text{bearing} = 90 + 33$$

$$= 123^\circ \text{ T.}$$

Done well

Question 10 (8 marks)

- (a) Susan and Nick are planning a fund-raising dance. They hire a hall for \$400 and a band for \$300. Refreshments will cost \$12 per person. They set the entry ticket at \$20 per person. The graph below shows planned income and costs for the dance.



- (i) Write a formula for the cost, \$C\$, of running the dance for x people. 1

$$C = 12x + 700$$

many did well.

- (ii) Estimate the minimum number of people needed at the dance to cover the costs. 1

87 people

many did well

- (iii) How much profit will be made if 150 people attend the dance? 1

$$P = I - C$$

$$P = 20 \times 150 - (12 \times 150 + 700)$$

$$P = \$500$$

many did well.

- (iv) Susan and Nick plan to sell 200 tickets. They want to make a profit of \$1 500. What should be the price of a ticket assuming all 200 tickets will be sold? 2

let the ticket price be x

many did well

$$1500 = 200x - (12 \times 200 + 700)$$

$$200x = 4600$$

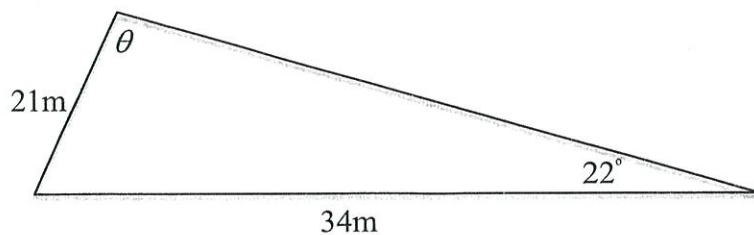
$$x = \frac{4600}{200}$$

$$x = \$23$$

Question 10 continues on next page

- (b) Calculate the value of θ , correct to the nearest minute.

3



NOT TO
SCALE

$$\frac{\sin \theta}{34} = \frac{\sin 22^\circ}{21}$$

$$\sin \theta = \frac{34 \sin 22^\circ}{21}$$

$$\theta = 37^\circ 20' \text{ or } 142^\circ 40'$$

- many forgot the
ambiguous case

- some forgot to change
the calculator mode
to degree.

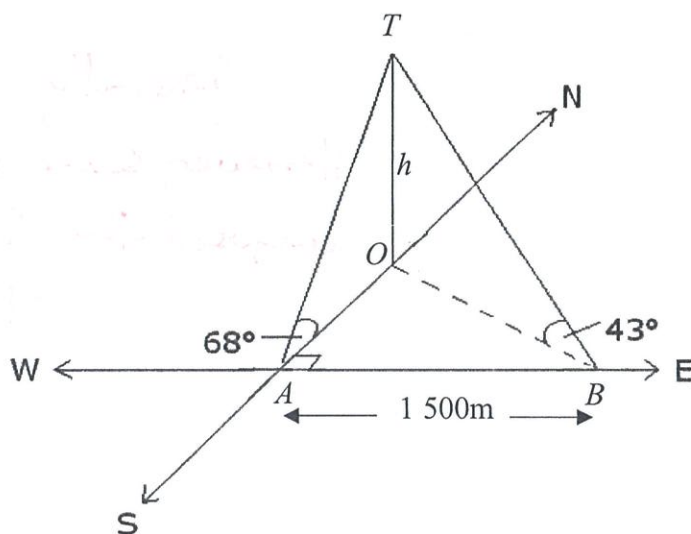
Question 11 (8 marks)

- (a) Solve for x : $\cos x = -\frac{\sqrt{2}}{2}$ for $0^\circ \leq x \leq 360^\circ$.

2

acute angle $x = 45^\circ$
 $\therefore QII = 180 - 45$ $QIII = 180 + 45^\circ$
 $\therefore x = 135^\circ$, $x = 225^\circ$ *well done!*

- (b) A lighthouse of height h metres, is observed from two points, A and B , in the same horizontal plane as the base of the lighthouse. From point A , due south of the lighthouse, the angle of elevation to the top of the lighthouse is 68° . From point B , due east of point A , the angle of elevation to the top of the lighthouse is 43° . Points A and B are 1 500 metres apart.



NOT TO
SCALE

- (i) Show that $OA = h \tan 22^\circ$

2

$\tan 68^\circ = \frac{h}{OA}$
 $OA = \frac{h}{\tan 68^\circ}$
 $OA = h \cot 68^\circ$ *well done!*
 $\therefore OA = h \tan 22^\circ$

Question 11 continues on next page

Question 11 (continued)

(ii) Show that $h = \frac{1500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}$

3

$$OB = h \tan 47^\circ \quad (\text{similarly } OA = h \tan 22^\circ)$$

$$\therefore 1500^2 = (h \tan 47^\circ)^2 - (h \tan 22^\circ)^2 \quad (\text{from Pythagoras Theorem})$$

$$1500^2 = h^2 (\tan^2 47^\circ - \tan^2 22^\circ)$$

$$h^2 = \frac{1500^2}{\tan^2 47^\circ - \tan^2 22^\circ}$$

$$\therefore h = \frac{1500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}, \quad h > 0$$

Generally well done!

However, some students confused the Pythagoras Theorem!

(iii) Find the height of the tower correct to the nearest metre.

1

$$\therefore h = 1399.514865 \dots$$

$$\therefore h = 1400 \text{ m (correct to nearest metre)}$$

Generally well done!

END OF PAPER